

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim : Simulate smoothing and sharpening operations on images using frequency domain filters.	
Experiment 9	Date: 23/08/2026	Enrolment No:92400133050

GitHub Link :

Aim

write a python program to simulate smoothening and sharpening operation on image using spatial filter

Code

```
import cv2

import numpy as np

from google.colab.patches import cv2_imshow

# Load the image

image = cv2.imread('/content/gratisography-moon-robot-800x525.jpg')

# Define the Gaussian kernel for smoothing

kernel_size = (5, 5)

sigma = 1.5

gaussian_kernel = cv2.getGaussianKernel(kernel_size[0], sigma)

gaussian_kernel = np.outer(gaussian_kernel, gaussian_kernel)

# Apply Gaussian smoothing

smoothed_image = cv2.filter2D(image, -1, gaussian_kernel)

# Define a sharpening kernel

sharpening_kernel = np.array([[ -1, -1, -1],
                               [ -1,  9, -1],
                               [ -1, -1, -1]])

# Apply sharpening

sharpened_image = cv2.filter2D(image, -1, sharpening_kernel)

# Display the original image, smoothed, and sharpened images

cv2_imshow(image)
```

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```
cv2_imshow(smoothed_image)
```

```
cv2_imshow(sharpened_image)
```

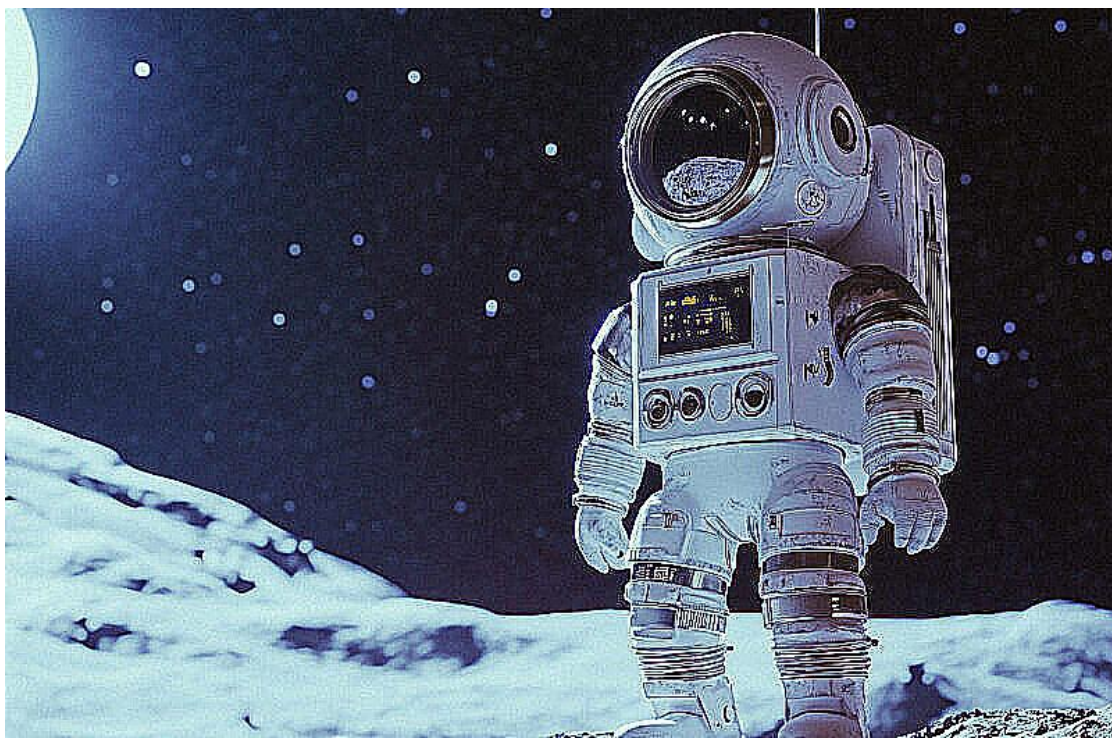
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Output



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Aim

write a python program to simulate smoothening using Averaging linear filters

Code

```
# write a python program to simulate smoothening using Averaging linear filters

import cv2

import numpy as np

from google.colab.patches import cv2_imshow

# Load the image

image = cv2.imread('/content/gratisography-moon-robot-800x525.jpg') # Changed image path

# Define the size of the Averaging filter kernel

kernel_size = (5, 5) # You can adjust the size based on the desired smoothing level

# Create the Averaging filter kernel

kernel = np.ones(kernel_size, dtype=np.float32) / (kernel_size[0] * kernel_size[1])

# Apply the Averaging filter for smoothing

smoothed_image = cv2.filter2D(image, -1, kernel)

# Display the original and smoothed images

cv2_imshow(image)

cv2_imshow(smoothed_image)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Output

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Aim

write a python program to simulate smoothening using median filters

Code

```
# write a python program to simulate smoothening using median filters

import cv2

import numpy as np

from google.colab.patches import cv2_imshow

# Load the image

image = cv2.imread('/content/gratisography-moon-robot-800x525.jpg') # Changed image path

# Define the size of the median filter kernel (should be an odd number)

kernel_size = 5 # You can adjust the size based on the desired smoothing level

# Apply the Median filter for smoothing

smoothed_image = cv2.medianBlur(image, kernel_size)

# Display the original and smoothed images

cv2_imshow(image)

cv2_imshow(smoothed_image)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Output

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Aim

write a python program to simulate sharpening using spatial high pass filters

Code

write a python program to simulate sharpening using spatial high pass filters

```
import cv2

import numpy as np

from google.colab.patches import cv2_imshow

# Load the image

image = cv2.imread('/content/gratisography-moon-robot-800x525.jpg') # Changed image path

# Apply Gaussian smoothing to reduce noise (optional but recommended)

blurred_image = cv2.GaussianBlur(image, (5, 5), 0)

# Create a Laplacian kernel for sharpening

laplacian_kernel = np.array([[0, -1, 0],
                             [-1, 5, -1],
                             [0, -1, 0]], dtype=np.float32)

# Apply the Laplacian filter for sharpening

sharpened_image = cv2.filter2D(blurred_image, -1, laplacian_kernel)

# Display the original image, blurred image, and sharpened image

cv2_imshow(image)

cv2_imshow(blurred_image)

cv2_imshow(sharpened_image)

cv2.waitKey(0)
```

Output

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Conclusion

The different spatial filters were successfully applied to the input image.

The Gaussian and averaging filters smooth the image and reduce noise.

The median filter also smooths the image and is useful for removing noise while preserving edges.

The sharpening and high-pass filters enhance edges and fine details in the image.

The results show that different filters produce different levels of smoothing and sharpening.

Thus, spatial filtering is useful for image enhancement, noise reduction, and edge/detail improvement.